

1. Information Retrieval

Information retrieval is the process of finding useful information from a collection in response to an information need. Search engines, digital libraries, enterprise search platforms, and retrieval-augmented generation systems all rely on retrieval concepts. A retrieval system receives a query, represents the query and available documents, calculates relevance, and returns ranked candidates. Traditional systems often depend on lexical matching, whereas semantic retrieval uses dense vector representations. Retrieval quality depends on document segmentation, indexing, representation, ranking, query formulation, and evaluation. The purpose is generally to place useful evidence near the top of the result list so that a downstream user or system can use it effectively.

2. Inverted Index

An inverted index is a fundamental data structure for lexical information retrieval. Instead of storing only complete documents, it maps each term to the documents or positions in which that term occurs. This allows a search engine to locate candidate documents efficiently without scanning every document for every query. Term frequency and document frequency statistics can then support ranking algorithms such as TF-IDF and BM25. Inverted indexes are particularly effective for exact words, technical identifiers, names, and phrases. Their main limitation is that lexical matching may fail when a relevant document expresses the same concept using different vocabulary. This limitation motivates semantic and hybrid retrieval systems.

3. BM25 Ranking

BM25 is a widely used lexical ranking function that estimates how relevant a document is to a query. It considers factors including query-term frequency, inverse document frequency, and document-length normalization. The method gives useful weight to terms that are distinctive within a collection while controlling the effect of very long documents. BM25 is often a strong baseline for information retrieval because it is efficient, interpretable, and effective for exact terminology. However, it mainly depends on lexical overlap. A query and relevant passage may be semantically related even when they share few words, in which case dense retrieval can complement BM25.

4. Dense Retrieval

Dense retrieval represents queries and documents as dense numerical vectors produced by an embedding model. Relevance can then be estimated using a similarity or distance measure between the query vector and document vectors. A vector database or approximate nearest-neighbor index can efficiently search large collections. Dense retrieval can recognize semantic relationships even when the exact query terms are absent from a passage. Its effectiveness depends on the embedding model, chunking strategy, domain, and retrieval configuration. A dense retriever is therefore best viewed as a candidate-generation component rather than a guarantee that the highest-ranked passage is the correct evidence.

5. Hybrid Search

Hybrid search combines multiple retrieval approaches, commonly lexical retrieval and dense semantic retrieval. Lexical retrieval is valuable for exact names, identifiers, codes, and rare technical terminology, while dense retrieval can identify passages with similar meaning despite vocabulary differences. Candidates from both systems can be merged, fused, or passed to a later reranking stage. Hybrid search introduces additional design decisions such as score normalization, result fusion, candidate limits, and weighting. It is particularly useful for technical or enterprise collections where users may alternate between precise keyword queries and broader conceptual questions.

6. Retrieval Evaluation

Retrieval evaluation measures whether a search system finds useful evidence and ranks it effectively. Precision at K measures how many of the top K results are relevant, while recall at K measures how much relevant material appears within those results. Ranking metrics can additionally account for the position of relevant items. In RAG, retrieval evaluation is crucial because the language model cannot reliably use evidence that the retriever never supplies. A retrieval test set should contain realistic queries and known relevant passages, along with difficult negative examples where possible. Logging queries, retrieved chunks, distances, relevance scores, and final selections can help identify whether failures originate from chunking, embeddings, indexing, or ranking.